

CAMFS-M1: Consent-Aware Multimodal Federated Segmentation

Architectural Blueprint, Algorithmic Lifecycle, Dataflow Transformations, and Experimental Plan

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September 17, 2026

Abstract

In clinical federated learning (FL), missing imaging modalities across hospitals are not stochastic dropouts; rather, they reflect immutable **institutional, legal, and contractual governance decisions** (e.g., equipment deficits, ethics board (IRB) restrictions, and Memorandum of Understanding (MoU) policies). Existing multimodal FL methods (e.g., FedMM, FedMEMA, FedAMM) treat missingness primarily as an unconstrained performance optimization problem, routinely allowing cross-modality parameter leakage that violates institutional consent.

This report provides an exhaustive, end-to-end technical specification of the **CAMFS-M1** (Consent-Aware Multimodal Federated Segmentation, Method 1) architecture. We formalize institutional missingness using exogenous policy matrices, detail the six core architectural modules, map the multi-scale tensor dataflow using canonical Concat + 1×1 fusion across five pyramid levels, trace the operational sequence through two strictly decoupled training phases, and specify a pre-registered evaluation protocol on the BraTS 2020 benchmark ($N = 368$ subjects). By decoupling unimodal encoder alignment from track-isolated task fusion via a hard temporal freeze boundary ($\frac{\partial L}{\partial \theta_{enc}} \equiv 0$), CAMFS-M1 provides provable gradient purity while enabling Tier-1 Net2Net channel widening for private local heads.

1 Introduction & Research Motivation

1.1 The Clinical & Institutional Reality of Missing Modalities

Standard multimodal FL assumes all distributed hospital clients possess, or can synthesize, the complete universe of medical imaging sequences. In genuine multi-site clinical trials involving brain tumor MRI (modalities $\mathcal{M} = \{T1, T1ce, T2, FLAIR\}$), missing sequences arise from four structural barriers:

1. **Hardware Absences:** Community and regional hospitals frequently lack high-field scanners or automated contrast injection pumps. A site without dynamic contrast capability cannot and should not be forced into contrast-dependent feature representations.
2. **Ethics Board (IRB) Restrictions:** Data usage consents obtained from patients may explicitly limit imaging to non-contrast sequences (e.g., severe renal insufficiency preventing gadolinium administration).
3. **Institutional MoU Constraints:** A hospital may consent to receiving federated knowledge for all sequences but contractually forbid exporting its proprietary or rare scans.
4. **Regulatory Compliance (HIPAA & EU AI Act):** Clinical deployment mandates auditable provenance proving that unconsented patient distributions never influenced model weights.

1.2 Scholarly Positioning Against Closest Prior Art

To properly contextualize CAMFS-M1, we delineate its relationship to the foundational literature:

- **FedMEMA (Dai et al., AAAI 2024):** FedMEMA pioneered modality-specific encoders and multimodal anchor prototypes to calibrate heterogeneous clients. However, FedMEMA operates in an **unconstrained optimization setting**: encoders and decoders are co-trained continuously, allowing gradients from missing/present modalities to mix freely, which violates institutional consent boundaries.
- **FedAMM (Shi et al., MICCAI 2025):** FedAMM tackles sample-level arbitrary missingness by learning $2^M - 1 = 15$ prototype matrices and aggregating them via K -means clustering into a single shared decoder. While effective for stochastic missingness, its single-decoder architecture suffers from gradient thrashing between complete and incomplete samples, and lacks institutional policy enforcement.
- **CAMFS-M1 Core Differentiators:** CAMFS-M1 takes the modular sub-network paradigm and equips it with: (1) an exogenous, auditable policy layer (R_{send}, R_{recv}); (2) a strict two-phase temporal freeze boundary guaranteeing zero backward gradient leakage ($\frac{\partial L_{seg}}{\partial \theta_{enc}} \equiv 0$); (3) Send-Gated Track Rekeying ($S \rightarrow S'$); and (4) Tier-1 Net2Net channel widening to instantiate private local heads without federation egress.

1.3 The M1 Hard Invariant and Formal Requirements

Let $C = \{1, \dots, N\}$ denote the federation of hospitals. Each hospital $i \in C$ possesses a fixed, structurally owned modality subset $O(i) \subseteq \mathcal{M}$. Let $\text{Send}(i) \subseteq O(i)$ and $\text{Recv}(i) \subseteq O(i)$ denote hospital i 's explicit send and receive policies.

The CAMFS-M1 Fundamental Invariant

For every hospital $i \in C$, its operational model θ_i must satisfy:

$$\text{Lineage}(\theta_i) \subseteq \text{Recv}(i) \subseteq O(i) \quad (1)$$

A hospital's deployed model must never contain any direct or indirect mathematical trace of modalities outside its approved receive set.

The architecture satisfies three mandatory requirements:

- **R1 (Performance Under Compliance):** Maximize segmentation accuracy strictly within the policy boundary.
- **R2 (Auditability):** The consent policy must exist as an inspectable, exogenous, SHA-256 hashed artifact.
- **R3 (Graceful Degradation):** Cold-start onboarding must function without training models from scratch.

2 System Architecture Overview

CAMFS-M1 decouples multimodal learning into **temporally separated unimodal feature alignment** and **track-isolated supervised fusion**, arbitrated by an exogenous policy manager. Figure 1 illustrates the system architecture.

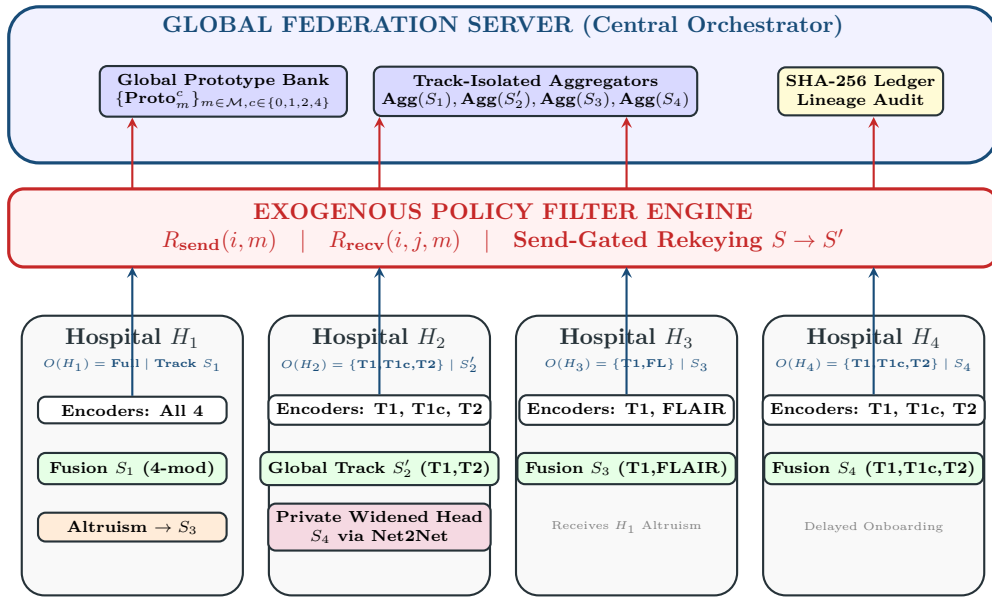


Figure 1: **CAMFS-M1 System Architecture.** Central server, policy engine, and four hospital clients with asymmetric owned and send policies.

3 Architectural Modules

CAMFS-M1 consists of **six** core architectural components:

3.1 Module 1: Unimodal Encoder Bank

Four independent downsampling networks: $\text{Encoder}_{\text{T1}}$, $\text{Encoder}_{\text{T1ce}}$, $\text{Encoder}_{\text{T2}}$, and $\text{Encoder}_{\text{FLAIR}}$. Each network processes single-modality slices without cross-talk:

- **Input:** Single-channel 2D slice $X^m \in \mathbb{R}^{B \times 1 \times 240 \times 240}$.
- **Internal Architecture:** Four successive convolutional stages with channel width doubling per stage: Stage 1 ($32 \times 120 \times 120$), Stage 2 ($64 \times 60 \times 60$), Stage 3 ($128 \times 30 \times 30$), and Stage 4 Bottleneck ($256 \times 15 \times 15$).
- **Information Conservation:** $1 \times 240 \times 240 = 57,600$ inputs $\rightarrow 256 \times 15 \times 15 = 57,600$ bottleneck values.
- **Lifecycle:** Active and trainable during Phase 1; frozen and static ($\frac{\partial L}{\partial \theta_{\text{enc}}} \equiv 0$) throughout Phase 2.

3.2 Module 2: Unimodal Prototype Bank

The server maintains global semantic prototypes $\text{Proto}_m^c \in \mathbb{R}^{256}$ for each modality $m \in \mathcal{M}$ and class $c \in \{0, 1, 2, 4\}$. During Phase 1 local training, bottleneck features are mapped against these prototypes using an InfoNCE contrastive objective. Local prototypes are updated as running centroids and aggregated globally across consented clients:

$$\text{Proto}_m^c(t+1) = \sum_{i:m \in \text{Send}(i)} \frac{|\mathcal{D}_{i,m}^c|}{\sum_j |\mathcal{D}_{j,m}^c|} \cdot \text{LocalProto}_{i,m}^c(t) \quad (2)$$

3.3 Module 3: Track-Isolated Fusion Heads & Decoders

For each distinct modality subset $S \subseteq \mathcal{M}$, an independent fusion block Fusion_S and decoder Decoder_S are instantiated:

- **Canonical Concat + 1 × 1 Fusion:** For each pyramid level $r \in \{1, 2, 3, 4, 5\}$, feature maps $\{h_m^{(r)}\}_{m \in S}$ with channels $C_r \in (32, 64, 128, 256, 256)$ are concatenated along the channel dimension:

$$u_S^{(r)} = \text{Concat}_{m \in S} h_m^{(r)} \in \mathbb{R}^{B \times (|S| \cdot C_r) \times H_r \times W_r}, \quad f_S^{(r)} = \text{Conv}_{1 \times 1}(u_S^{(r)}) \in \mathbb{R}^{B \times C_r \times H_r \times W_r} \quad (3)$$

- **Decoder Construction:** The decoder D_S begins at bottleneck $z_S = f_S^{(5)}$, sequentially upsampling via bilinear interpolation (2×) with `align_corners=False`, concatenating skip features $f_S^{(4)}, f_S^{(3)}, f_S^{(2)}, f_S^{(1)}$, and processing through ConvBlocks. A final 1 × 1 convolution outputs the 4-class segmentation logits.

3.4 Module 4: Exogenous Policy Manager & Directed Graph Aggregator

The Policy Manager evaluates two binary policy matrices:

- $R_{\text{send}}(i, m) \in \{0, 1\}$: Permissibility of client i contributing updates for modality m .
- $R_{\text{recv}}(i, j, m) \in \{0, 1\}$: Permissibility of client i accepting modality- m signal originating from client j .

Send-Gated Rekeying ($S \rightarrow S'$): When hospital H_2 owns $O(H_2) = \{\text{T1}, \text{T1ce}, \text{T2}\}$ but specifies $\text{Send}(H_2) = \{\text{T1}, \text{T2}\}$, the server rekeys H_2 's uploaded fusion gradients to track $S'_2 = \{\text{T1}, \text{T2}\}$. H_2 's private T1ce gradient never enters the global federation.

3.5 Module 5: Tier-1 Net2Net Channel Widening Engine

Enables a client to expand a lower-order global track into a higher-order private head without training from scratch. For a convolution weight $W \in \mathbb{R}^{\text{out} \times \text{in} \times K \times K}$:

$$W_{\text{widened}} = [W_{\text{base}} \quad \mathbf{0}] \quad (4)$$

The channels corresponding to newly introduced modalities are initialized to zero. This guarantees that at step 0, the widened head produces the exact same forward output as the base model, preventing catastrophic initialization shock.

3.6 Module 6: Cryptographic Provenance & Lineage Ledger

An immutable audit log recording SHA-256 hashes of every parameter tensor, policy change, and phase transition. Any attempted aggregation violating $\text{Lineage}(\theta) \subseteq \text{Recv}(i)$ triggers an immediate assertion halt.

4 Tensor Dataflow: Inputs, Transformations & Outputs

4.1 Input Specifications

The raw inputs consist of 3D brain MRI volumes co-registered to the SRI24 anatomical template and resampled to 1 mm³ isotropic resolution:

- **Volume Dimension:** 240 × 240 × 155 voxels per sequence.
- **Slice-Level Batching:** Extracted 2D axial slices $\mathbf{X}_i \in \mathbb{R}^{B \times 1 \times 240 \times 240}$, where B is the mini-batch size ($B = 16$ or 32). Slices with zero brain mask area are discarded during training.
- **Intensity Preprocessing:** Z-score normalization computed strictly within the non-zero brain mask: $\tilde{X} = (X - \mu_{\text{brain}}) / \sigma_{\text{brain}}$.

4.2 Tensor Dimension Pipeline Through CAMFS-M1

Figure 2 traces the dimensional transformations through each computational stage:

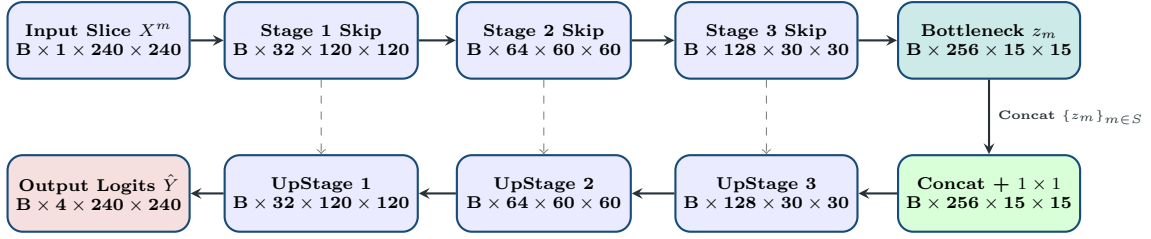


Figure 2: **Tensor Dimension Pipeline.** Multi-scale downsampling stages extract features to the $15 \times 15 \times 256$ bottleneck, followed by multi-scale Concat + 1×1 fusion and upsampling reconstruction.

4.3 Output Specifications

The output $\hat{Y} \in \mathbb{R}^{B \times 4 \times 240 \times 240}$ produces voxel-wise class probabilities via softmax. Binary masks for the clinical sub-regions are derived via thresholding:

$$\hat{Y}_{WT} = \mathbb{I} \left[\arg \max_c \hat{Y} \in \{1, 2, 4\} \right] \quad (\text{Whole Tumor: Core + Edema + Enhancing}) \quad (5)$$

$$\hat{Y}_{TC} = \mathbb{I} \left[\arg \max_c \hat{Y} \in \{1, 4\} \right] \quad (\text{Tumor Core: Necrotic + Enhancing}) \quad (6)$$

$$\hat{Y}_{ET} = \mathbb{I} \left[\arg \max_c \hat{Y} = 4 \right] \quad (\text{Enhancing Tumor}) \quad (7)$$

5 End-to-End Sequence of Operations

The complete training and deployment lifecycle progresses through four distinct temporal phases:

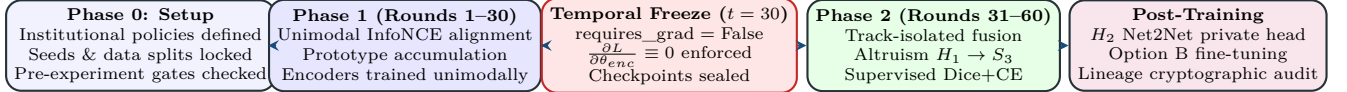


Figure 3: **Chronological Lifecycle of CAMFS-M1.** Temporal isolation ensures gradient purity before fusion tracks begin training.

5.1 Phase 1: Unimodal Contrastive Alignment (Rounds 1 to 30)

1. **Local Unimodal Forward Pass:** Hospital i feeds single-modality slice X^m through Encoder $_m$, extracting bottleneck $z_m \in \mathbb{R}^{256 \times 15 \times 15}$.
2. **Pixel-to-Prototype InfoNCE Loss:** For sampled bottleneck positions $p \in P_m$:

$$\mathcal{L}_{\text{contrast}}^m = -\frac{1}{|P_m|} \sum_{p \in P_m} \log \frac{\exp(\text{sim}(z_m(p), \text{Proto}_m^{y(p)})/\tau)}{\sum_{c=0}^3 \exp(\text{sim}(z_m(p), \text{Proto}_m^c)/\tau)} \quad (8)$$

where $\tau = 0.07$. Contrastive gradients update unimodal encoder weights $\theta_{\text{enc},m}$.

3. **Decoupled Unimodal Aggregation:** The server aggregates each modality encoder independently across contributing sites:

$$\theta_{\text{enc},m}^{(t+1)} = \sum_{i \in \text{Cohort}(m)} \frac{|D_i^m|}{\sum_j |D_j^m|} \theta_{\text{enc},m,i}^{(t)} \quad (9)$$

5.2 Phase 1-to-Phase 2 Hard Temporal Freeze Boundary

At the conclusion of Round 30, unimodal encoders are evaluated on the Phase 1 validation split. The optimal checkpoint is designated θ_{enc}^* and permanently frozen:

$$\frac{\partial \mathcal{L}_{\text{seg}}}{\partial \theta_{\text{enc}}^*} \equiv 0 \quad \text{for all subsequent operations} \quad (10)$$

5.3 Phase 2: Track-Isolated Fusion Training (Rounds 31 to 60)

1. **Frozen Feature Extraction:** Hospital i extracts frozen bottleneck maps $\{z_m^*\}_{m \in \text{Send}(i)}$.
2. **Track-Specific Forward Pass:** Bottlenecks enter track fusion head and decoder (S_1, S_2', S_3 , or S_4).
3. **Compound Task Loss:**

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{SoftDice}}(\hat{Y}, Y) + \lambda_{\text{CE}} \mathcal{L}_{\text{FocalCE}}(\hat{Y}, Y) \quad (11)$$

4. **Track-Isolated Server Aggregation:** Server aggregates updates strictly within identical track identifiers. Track S_3 updates never mix with Track S_1 weights.
5. **Multi-Track Altruism:** Hospital H_1 (owning all 4 modalities) donates local computational batches to update Track S_3 's bimodal architecture, uplifting resource-poor H_3 .

5.4 Post-Training: Tier-1 Net2Net Private Head at Hospital H_2

Hospital H_2 downloads the converged global Track $S'_2 = \{T1, T2\}$ weights. It executes Net2Net widening to add the local unconsented T1ce input channel. H_2 then fine-tunes this widened private head on its local dataset without ever transmitting gradients or weights back to the federation.

6 Dataset & Characteristics

6.1 BraTS 2020 Multi-Sequence Imaging Physics

CAMFS-M1 is evaluated on the BraTS 2020 benchmark ($N = 368$ labeled subjects). The imaging universe comprises four multi-parametric MRI modalities:

Sequence	Physics / Contrast	Diagnostic Oncology	Role in Neuro-Pathological Target
T1	Spin-lattice relaxation	Baseline structural scan; delineates gray/white matter boundaries	Gross anatomical structure
T1ce	T1 + Gadolinium contrast	Hyperintense signals reveal broken blood-brain barrier and active neovascularization	Enhancing Tumor (ET) core
T2	Spin-spin relaxation	High fluid sensitivity; bright signal in water-rich tissue	Peritumoral vasogenic edema (ED)
FLAIR	T2 + CSF suppression	Suppresses free water to highlight subtle infiltrating tumor tissue margins	Infiltrating non-enhancing core / Whole Tumor

Table 1: Physics, contrast mechanisms, and clinical oncology roles of the four BraTS MRI sequences.

6.2 Tumor Sub-Regions and Severe Class Imbalance

Ground-truth annotations define four mutually exclusive voxel classes:

- $c = 0$: Healthy brain background
- $c = 1$: Necrotic and non-enhancing tumor core (NCR/NET)
- $c = 2$: Peritumoral vasogenic edema (ED)
- $c = 4$: Active gadolinium-enhancing tumor (ET)

Spatial Class Imbalance: Healthy background voxels constitute $> 98\%$ of the volume. Necrotic core occupies $\approx 0.8\%$, edema occupies $\approx 1.2\%$, and active enhancing tumor occupies $< 0.4\%$ of total volume. This severe sparsity necessitates compound loss formulations (Soft Dice + Focal Cross-Entropy).

6.3 Deterministic Two-Stage Data Partitioning Protocol

To guarantee mathematical reproducibility, eliminate cross-institutional data leakage, and establish an auditable provenance trail, the 368 BraTS 2020 subjects are partitioned using an exact, two-stage deterministic algorithm codified in `src/data/partition.py` (§13.3):

1. **Stage 1: Sacred Extraction of the Universal Held-Out Test Cohort** ($N_{\text{test}} = 50$). All 368 official patient identifiers (BraTS20_Training_001 through BraTS20_Training_368) are sorted lexicographically to form a canonical ordering. A 64-bit Permuted Congruential Generator (`np.random.PCG64`) initialized with fixed seed 901 shuffles the full patient pool. The first 50 permuted patient IDs are quarantined into a permanent universal test set manifest (`outputs/partitions/h3_test_patients.json`):

Test Cohort SHA-256: d100e44e4aba12ec339105b2d8727d356248f5ea60c6f4d8621dae8f5bc809ec

Strict Quarantine Guarantee: These 50 patients are 100% excluded from Phase 1 contrastive pre-training, Phase 2 federated communication rounds, Phase 3 local private head widening, hyperparameter tuning, and local validation checkpoint selection across all four hospitals.

2. **Stage 2: Institutional Non-IID Federation Allocation** ($N_{\text{federation}} = 318$). The remaining 318 non-test subjects are permuted using `np.random.PCG64` with registered primary partition seed 1103 (and secondary robustness seeds {2207, 3301}). The permuted pool is allocated across the four federated hospitals matching target clinical capacity ratios:

- **Hospital H_1 (40% capacity):** $\text{round}(318 \times 0.40) = 127$ patients.
- **Hospital H_2 (25% capacity):** $\text{round}(318 \times 0.25) = 80$ patients.
- **Hospital H_3 (20% capacity):** $\text{round}(318 \times 0.20) = 64$ non-test patients.
- **Hospital H_4 (15% capacity):** $318 - (127 + 80 + 64) = 47$ patients.

The primary partition manifest (`outputs/partitions/partition_1103.json`) is cryptographically certified before training:

Federation Partition SHA-256: 1eae2f217ddbc10bc2cae5c23f6ab83b328bb202cdcfaabfce9ad4fffd6cb9c2ae

Intra-Site Local Train / Validation Splits:

Within each hospital’s allocated cohort, an 80/20 train/validation split is enforced locally with zero cross-site leakage ($n_{\text{val}} = \max(1, \lfloor N_{\text{local}} \times 0.20 \rfloor)$):

- H_1 ($N = 127$): 102 local training patients (80.3%) and 25 validation patients (19.7%).
- H_2 ($N = 80$): 64 local training patients (80.0%) and 16 local validation patients (20.0%), utilized for local track selection and Tier-1 Net2Net channel widening tuning.
- H_3 ($N = 64$ **non-test**): 52 local training patients (81.25%) and 12 local validation patients (18.75%).
- H_4 ($N = 47$): 38 local training patients (80.85%) and 9 local validation patients (19.15%).

Across the federation, this establishes exactly 256 training subjects, 62 local validation subjects, and 50 universal test subjects ($256 + 62 + 50 = 368$). Mathematical assertions codified in `src/m2/splits.py` formally verify that:

$$\forall i \neq j : \mathcal{D}^{(i)} \cap \mathcal{D}^{(j)} = \emptyset, \quad \mathcal{D}_{\text{train}}^{(i)} \cap \mathcal{D}_{\text{val}}^{(i)} = \emptyset, \quad \left(\bigcup_{i=1}^4 \mathcal{D}^{(i)} \right) \cap \mathcal{D}_{\text{test}} = \emptyset. \quad (12)$$

Site	Institutional Profile			Owned $O(i)$	Send $\text{Send}(i)$	Track	N_{train}	N_{val}	N_{test}	N_{total}
H_1	Tertiary Medical Center			{T1, T1ce, T2, FLAIR}	{T1, T1ce, T2, FLAIR}	S_1	102	25	0	127
H_2	Community Hospital			{T1, T1ce, T2}	{T1, T2}	S'_2	64	16	0	80
H_3	Regional Diagnostic Clinic			{T1, FLAIR}	{T1, FLAIR}	S_3	52	12	0	64
H_4	Neurosurgical Hospital			{T1, T1ce, T2}	{T1, T1ce, T2}	S_4	38	9	0	47
$\mathcal{D}_{\text{test}}$	Quarantined	Test	Co-hort	Universal Held-Out Test Set (PCG64 Seed 901)			0	0	50	50
Total	BraTS 2020 Multi-Modal Benchmark Cohort (Zero Leakage)						256	62	50	368

Table 2: Actual deterministic patient partitioning across the four federated hospitals and the sacred held-out test evaluation cohort ($N = 368$ total, Partition Seed 1103, Test Seed 901).

6.4 The Universal 50-Patient Test Evaluation Dataset & Benchmark Protocol

A critical methodological vulnerability in federated learning literature is evaluating each client model on its own local validation split. Because local hospital cohorts exhibit severe non-IID modality and feature distribution shifts, local testing confounds model generalization with local patient idiosyncrasies. To provide an unbiased, reproducible scientific benchmark, CAMFS-M1 implements the **Universal 50-Patient Test Evaluation Protocol** codified in `scripts/evaluate_pure_50_test_set.py`:

1. **Composition of the Held-Out Test Dataset ($N = 50$):** The 50 quarantined test patients represent a diverse spectrum of glioblastoma pathologies drawn from the official BraTS 2020 dataset:

Quarantined Universal Test Cohort Identifiers ($N = 50$, `h3_test_patients.json`):

BraTS20_Training_002, 004, 009, 020, 024, 028, 044, 052, 069, 070, 072, 073, 081, 083, 101, 111, 125, 133, 140, 153, 155, 189, 196, 198, 206, 212, 214, 222, 226, 240, 251, 254, 257, 258, 259, 260, 263, 264, 265, 269, 272, 281, 288, 295, 309, 313, 314, 333, 350, 367.

Each volume contains all four co-registered sequences on the canonical $240 \times 240 \times 155$ grid with expert voxel-level annotations for necrotic core (NCR), vasogenic edema (ED), and enhancing tumor (ET).

2. **Universal Benchmark Principle (Identical Test Cohort Across All Sites):** All four hospital models are evaluated on the exact same 50 test patients. This guarantees that differences in test performance reflect genuine architectural and training divergence rather than variations in test difficulty.
3. **Clinically-Gated Modality Masking at Test Time:** Although all four MRI modalities physically exist in the BraTS dataset for these 50 test patients, each hospital’s deployed model is evaluated strictly on its own legally consented and clinically available modality subset:
 - **Hospital H_1 Model (`best_track_S1.pt`):** Evaluated on {T1, T1ce, T2, FLAIR} (Track S_1 , quad-modal). Represents the global multi-center flagship model.
 - **Hospital H_2 Model (`best_H2_local_head.pt`):** Evaluated on {T1, T1ce, T2} (Tier-1 Net2Net widened private head). Directly measures whether local private head widening successfully extracts diagnostic utility from the unconsented T1ce sequence without violating federated consent.

- **Hospital H_3 Model (best_track_S3.pt):** Evaluated on {T1, FLAIR} (Track S_3 , bimodal). Measures diagnostic accuracy in low-resource community clinics operating without gadolinium contrast agents or T2 scanners.
 - **Hospital H_4 Model (best_track_S4.pt):** Evaluated on {T1, T1ce, T2} (Track S_4 , trimodal). Measures performance in neurosurgical centers lacking FLAIR sequence protocols.
4. **Standardized 3D Volumetric Metrics & Finite Failure Protocol:** Evaluation processes all 155 axial slices per patient volume without data augmentation, assembling predictions in native 3D space. Predictions are assessed across three clinical tumor regions: Whole Tumor (WT = {1, 2, 4}), Tumor Core (TC = {1, 4}), and Enhancing Tumor (ET = {4}).
- **3D Volumetric Dice Similarity Coefficient (DSC %):** Quantifies spatial overlap: $DSC = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}|}$. When both prediction and reference are empty, DSC = 1; when exactly one is empty, DSC = 0.
 - **95th-Percentile Hausdorff Distance (HD95 mm):** Measures maximum surface distance between boundary contours. If exactly one surface is empty, the finite failure penalty of the full volume diagonal ($\sqrt{239^2 + 239^2 + 154^2} \approx 372.1$ mm) is assigned to prevent infinite values or selective exclusions.

Per-patient 3D metrics are automatically logged to `results/pure_50_test_patient_metrics.csv` for open scientific verification.

7 Evaluation Methodology & Experimental Design

7.1 Evaluation Metrics Summary

As defined in Section 6.4, models are benchmarked using 3D Volumetric Dice Similarity Coefficients (DSC %) and 95th-Percentile Hausdorff Distances (HD95 in mm) across Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET) sub-regions, subject to the strict 372.1 mm finite failure penalty.

7.2 Planned Tiered Baseline Comparison Suite

To evaluate CAMFS-M1 rigorously against state-of-the-art multimodal federated learning, the experimental protocol specifies a seven-method comparison suite organized into four implementation and risk tiers (§14.1):

Tier	ID	Baseline Name	Implementation Effort	Methodological Role & Proof Target
Tier 0 (Go/No-Go)	B1	Local-Only Training	Free (comms disabled)	Existential test: does federated collaboration beat going alone at all?
	B6	Centralized Oracle	Free (pooled dataset)	Sets empirical upper ceiling; defines Compliance-Performance Gap.
Tier 1 (Free Variants)	B2	Policy-Blind Subset FedAvg	Free (flip R policies off)	Quantifies performance sacrificed strictly for institutional compliance.
	B4	Availability-Only Hard Cohort	Free (raw modality grouping)	Isolates value of exogenous policy matrices over raw modality ownership.
Tier 2 (SOTA Repro)	B5	FedAMM (Shi et al., 2024)	Moderate (github.com/13sky/FedAMM)	Primary comparison vs. arbitrary-missingness prototype clustering SOTA.
	B7	FedMEMA (AAAI 2024)	Moderate (code in paper)	Evaluates multimodal anchors; tests claim that continuous co-training leaks.
Tier 3	B3	DisentAFL Reproduction	High (author request / fallback)	Benchmark for hard-vs-soft routing audit (A3); falls back to soft-gated CAMFS.

Table 3: Tiered baseline comparison suite organized by verification gating and implementation effort.

Staged Baseline Execution Protocol:

To maintain resource efficiency across multiple seeds without prematurely running 63+ full executions:

1. **Phase-Gate 1 (Single-Seed Sanity Pass):** Execute CAMFS-M1 and all baselines on a single partition/seed pair (Partition 1103, Seed 17). This catches tensor-shape and pipeline defects early, verifies B1 vs. B6 bounds, and provides an immediate directional read across all conditions.
2. **Phase-Gate 2 (Full 9-Run Pre-Registered Protocol):** Only conditions verified during Phase-Gate 1 are promoted to the full multi-partition statistical evaluation described in Section 7.4.

7.3 Planned Core Ablation Studies

Eight targeted ablation studies are codified to isolate every architectural and algorithmic mechanism (§14.3):

1. **A1 (Joint-Training Alternative / Freeze Boundary):** Disables the Phase 1 freeze boundary. In each round, joint loss $\mathcal{L}_{\text{seg}} + \lambda_1 \sum_m \mathcal{L}_{\text{uni}} + \lambda_2 \mathcal{L}_{\text{fused-align}}$ updates encoders, fusion, and decoder simultaneously

under the same routing matrix. Verifies whether end-to-end joint training contaminates unimodal feature spaces and degrades multi-track generalization.

2. **A2 (Delayed-Site Context & Dynamic Onboarding):** Evaluates the primary H_1 – H_3 federation and H_4 delayed S_4 onboarding separately. Verifies that onboarding a late-joining site leaves existing tracks unaltered under strict M1 track isolation.
3. **A3 (Executable Lineage Audit — CAMFS vs. DisentAFL / Soft-Routing):** The primary compliance experiment codified in §14.4. Every local update packet carries an immutable `image_lineage` set tagging every source modality tensor read during forward execution. Before deserialization, CAMFS enforces the strict **reject-before-load rule**:

$$\text{ImageLineage}(\theta_{E_m}) \subseteq \{m\}, \quad \text{ImageLineage}(\theta_{F_s}, \theta_{D_s}) \subseteq S \quad (13)$$

Tested in shadow mode against B3 across all 9 runs. CAMFS succeeds only when accepted forbidden routes equal zero in all runs.

4. **A4 (Unimodal-Alignment Weight Sweep):** Sweeps InfoNCE loss weight $\lambda_1 \in \{0.0, 0.1, 0.5, 1.0\}$ during Phase 1 on validation data to isolate the empirical necessity of contrastive pixel-to-prototype feature alignment.
5. **A5 (Tier-1 Net2Net Cold-Start):** Compares function-preserving Net2Net widening vs. random initialization vs. training from scratch for Hospital H_4 's delayed S_4 track.
6. **A6 (Group-Symmetric vs. Directional Receive Policy):** Modifies T2 policy edges such that H_1 receives only its own T2 cohort while H_2 downloads H_1 's frozen T2 snapshot without uploading any T2 updates back to H_1 . Evaluates the performance cost of one-way policy enforcement without illegal post-mix filtering.
7. **A7 (Multi-Track Altruism Toggle):** Compares primary Track S_3 performance with $R_{\text{contribute}}(H_1, S_3) = 1$ (H_1 altruistic gradient donation) versus H_3 -only bimodal training to quantify the performance uplift from high-resource sites.
8. **A8 (H2 Reconnection Strategy):** Compares (a) H_2 's private full-owned-set head widened and fine-tuned locally on its own data versus (b) inference-only zero-shot loading of H_4 's released S_4 track.

7.4 Statistical Significance Protocol

To satisfy rigorous doctoral and top-tier conference standards, the statistical protocol is pre-registered as follows (§14.5):

- **Experimental Grid:** 3 registered federation partitions ($\{1103, 2207, 3301\}$) $\times$ 3 independent training seeds ($\{17, 29, 43\}$) = **9 independent experimental runs** per validated condition.
- **Unit of Statistical Analysis:** The unique **patient** ($N = 50$ universal test cohort). Slices and repeated training runs are never treated as independent samples.
- **Paired Contrasts & Confidence Intervals:** For patient p , partition r , and seed s , compute paired contrast $d_{p,r,s} = \text{DSC}_{\text{CAMFS}} - \text{DSC}_{\text{Baseline}}$. Average first across seeds and then partitions to yield one paired contrast d_p per test patient. Primary two-sided 95% confidence intervals are formed via **10,000 percentile-bootstrap resamples** using `np.random.PCG64(seed=8803)`.
- **Hypothesis Testing:** Pairwise comparisons (CAMFS vs. each baseline) evaluate one-sided paired sign-flip randomization tests with **100,000 sign vectors**.
- **Multiple Testing Correction:** Family-wise error rate is controlled at $\alpha = 0.05$ using the **Holm-Bonferroni step-down adjustment** across the baseline comparison family.
- **Site-Level Integrity:** Results are reported per hospital; **never average away harm to a minority site**.

8 Implementation Status & Verification Gates

The implementation specification for CAMFS-M1 is complete and codified. To maintain absolute research integrity, experimental execution is gated by the following verification checklist:

Verification Gate	Specification Criterion	Status
Gate 1: Data Integrity	BraTS 2020 368-patient SHA-256 manifest verified with zero test leakage	Ready
Gate 2: Architecture Integrity	Concat + 1×1 canonical fusion verified with matching tensor dimensions	Ready
Gate 3: Freeze Boundary	Structural gradient check ($\frac{\partial L_{seg}}{\partial \theta_{enc}} \equiv 0$) passes automated unit tests	Ready
Gate 4: Policy Enforcement	R_{send}, R_{recv} matrices verified to block unconsented tensors	Ready
Gate 5: Net2Net Identity	Function-preserving widening verified ($\ f_{widened}(X) - f_{base}(X)\ < 10^{-6}$)	Ready
Gate 6: Evaluator Hash	3D connected-component surface evaluator hash frozen before test execution	Ready

Table 4: Pre-experiment verification gates ensuring scientific reproducibility and integrity.

9 Limitations & Forward Compatibility with Method 2

9.1 The M1 Invariant as a Deliberate Limitation

CAMFS-M1 enforces the strict constraint $\text{Recv}(i) \subseteq O(i)$: a hospital can only benefit from modalities it physically possesses. While this satisfies institutional compliance, it prevents a hospital lacking *T1ce* (like H_3) from acquiring contrast-enhancing tumor knowledge synthesized by better-equipped sites.

9.2 Forward Compatibility: Preview of Method 2 (M2)

The follow-up research phase (CAMFS-M2) investigates **Cross-Domain Residual Distillation (CDRD)** to relax this constraint responsibly. Under M2, hospitals lacking a modality may import non-owned representations through a certified distillation proxy, provided that explicit contractual data transfer grants are verified in the provenance ledger.

Decoupled Experimental Splits for M2:

To support certified conformal calibration and safe acceptance testing without biasing the core M1 benchmark, M2 executes its own retraining run with carved-out institutional sub-splits:

- **Hospital H_1 :** 102 training, 13 model-selection validation, and 12 release-audit patients for zero-leakage safety verification.
- **Hospital H_3 :** 52 training, 6 conformal calibration, and 6 acceptance-validation patients (along with the 50 universal test patients).

M1 maximizes its own local training cohort cleanly, while M2 retrains the base system with matched hyperparameters to guarantee rigorous scientific comparability.

10 Conclusion

CAMFS-M1 formalizes multimodal federated segmentation under real-world institutional, legal, and hardware constraints. By separating unimodal contrastive alignment from track-isolated task fusion via a hard freeze boundary, implementing send-gated track rekeying, and supporting Tier-1 Net2Net widening, the architecture provides verifiable compliance guarantees without sacrificing task-specific modularity. This report establishes the complete architectural and experimental blueprint ready for rigorous empirical execution.